LSA (Latent Semantic Analysis) Recommender

After building the TF-IDF baseline, I extended the recommender with **Latent Semantic Analysis (LSA)** using **Truncated SVD**. The idea is to start from the same TF-IDF matrix, but then reduce that sparse high-dimensional representation into a smaller latent space.

Now the definition of similarity changes. In the TF-IDF baseline, songs are similar when they share many of the same distinctive words or phrases. In the LSA model, songs are similar when they occupy nearby positions in a lower dimensional semantic space shaped by broader word co-occurrence patterns across the corpus.

We can then capture songs expressing similar themes without using the exact same vocabulary. By compressing the TF-IDF matrix into latent dimensions, LSA can capture relationships that are not always visible through direct lexical overlap alone.

1. Imports & Setup

In [1]:
```
import sys
sys.path.append('../src')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.decomposition import TruncatedSVD
from sklearn.feature_extraction.text import TfidfVectorizer

from lsa_recommender import (
    build_lsa_recommender,
    recommend_songs,
    get_top_topics,
    get_song_top_topics
)
# Import TF-IDF recommender for direct side-by-side comparison
from tfidf_recommender import (
    build_tfidf_recommender,
    recommend_songs as tfidf_recommend
)

pd.set_option('display.max_colwidth', 60)
```

2. Recommenders

In [2]:
```
LYRICS_PATH = '../data/processed/lyrics_clean.csv'
N_COMPONENTS = 100  # number of latent topics - to be experimented with belo

# LSA
lyrics_df, lsa_matrix, lsa_sim, svd, vectorizer = build_lsa_recommender(
    LYRICS_PATH, n_components=N_COMPONENTS
)

# TF-IDF baseline
_, tfidf_matrix, tfidf_sim = build_tfidf_recommender(LYRICS_PATH)

print(f'Songs:              {len(lyrics_df)}')
print(f'TF-IDF matrix:      {tfidf_matrix.shape}  (songs x vocab terms)')
print(f'LSA matrix:         {lsa_matrix.shape}   (songs x latent topics)')
print(f'Explained variance: {svd.explained_variance_ratio_.sum():.1%} of tot
```

```
Songs:              1502
TF-IDF matrix:      (1502, 10000)  (songs x vocab terms)
LSA matrix:         (1502, 100)   (songs x latent topics)
Explained variance: 22.4% of total variance captured in 100 components
```

The original TF-IDF matrix has shape **(1502, 10000)**, meaning that 1,502 songs are represented across 10,000 vocabulary features. After applying Truncated SVD with **100 components**, the representation is reduced to an **LSA matrix of shape (1502, 100)**.

This reduced representation captures approximately **22.4% of the total variance** in the original TF-IDF space. While that means some information is compressed away, the tradeoff is that the model can focus on broader latent structure instead of isolated overlap between words alone. The LSA recommender is designed to sacrifice some surface detail in exchange for a more general semantic representation of lyrical similarity.

3. Inspecting the Latent Topics

Each component is defined by terms that tend to co-occur across songs, allowing the model to group together songs that may share themes, moods, or narrative structure even when their exact wording differs. Inspecting the highest weighted terms in each component helps show what kinds of lyrical patterns the recommender is learning from the corpus.

In [3]:
```
topics_df = get_top_topics(svd, vectorizer, n_topics=20, n_terms=8)
print("Top 8 terms per latent topic:\n")
topics_df
```

```
Top 8 terms per latent topic:
```

Out[3]:

	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9	Topic 10	Topic 11	Top 1
0	im	la	ooh	yeah	oh	dont	love	oh	im	hey	que	he
1	dont	la la	ooh ooh	yeah yeah	yeah	know	baby	oh oh	baby	hey hey	im	wanr
2	know	que	yeah	nigga	oh oh	wanna	dont	baby	hey	que	el	he he
3	like	ooh	oh	like	yeah yeah	dont know	want	im	hey hey	know	te	do wanr
4	just	el	yeah yeah	bitch	baby	dont wanna	love love	nigga	la la	baby	love	do
5	love	doo	oh oh	got	wanna	know know	hey	bitch	gonna	want	en	i
6	oh	en	love	im	dont	im	baby baby	got	im gonna	know know	lo	
7	yeah	ooh ooh	ah	fuck	know	want	wanna	like	yeah	el	mi	oh c

The latent topics still contain heavy overlap in adlibs, repeated phrases, and common emotional words, which suggests that LSA is smoothing the TF-IDF space but not fully escaping the lexical biases of the corpus. Topic structure is therefore partially semantic, but still strongly shaped by repeated lyric fragments and genre specific filler.

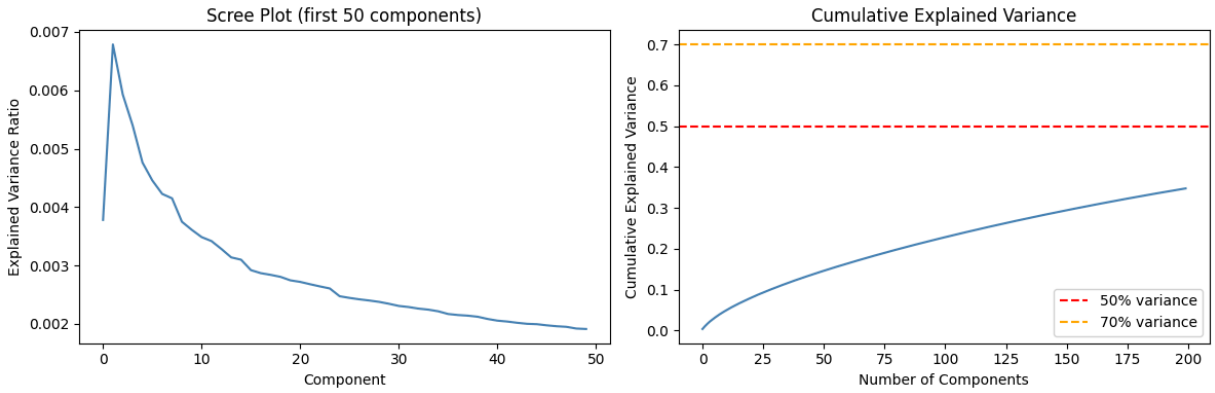
Scree Plot for Choosing n-components

How much variance does each additional component add? The "elbow" in this plot is a natural stopping point.

In [3]:

```
vec_inspect = TfidfVectorizer(max_features=10000, ngram_range=(1, 2), min_df
tfidf_inspect = vec_inspect.fit_transform(lyrics_df['processed_lyrics'])

svd_inspect = TruncatedSVD(n_components=200, random_state=42)
svd_inspect.fit(tfidf_inspect)

cumvar = svd_inspect.explained_variance_ratio_.cumsum()

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].plot(svd_inspect.explained_variance_ratio_[:50], color='steelblue')
axes[0].set_xlabel('Component')
axes[0].set_ylabel('Explained Variance Ratio')
axes[0].set_title('Scree Plot (first 50 components)')

axes[1].plot(cumvar, color='steelblue')
axes[1].axhline(0.5, color='red', linestyle='--', label='50% variance')
axes[1].axhline(0.7, color='orange', linestyle='--', label='70% variance')
```

```
axes[1].set_xlabel('Number of Components')
axes[1].set_ylabel('Cumulative Explained Variance')
axes[1].set_title('Cumulative Explained Variance')
axes[1].legend()

plt.tight_layout()
plt.show()

for threshold in [0.3, 0.5, 0.7]:
    n = (cumvar >= threshold).argmax() + 1
    print(f'{threshold:.0%} variance captured at {n} components')
```




```
30% variance captured at 156 components
50% variance captured at 1 components
70% variance captured at 1 components
```

The LSA model reduced the 10,000 dimensional TF-IDF space to 100 latent dimensions and captured 22.4% of the variance. That is enough to smooth the sparse TF-IDF space and reveal broader co-occurrence structure, but it also means a large amount of information remains outside the reduced space. In practice, this tradeoff makes recommendations less dependent on exact wording, but sometimes more prone to connecting songs that are only weakly related.

The printed result that 50% and 70% variance were reached at 1 component should not be interpreted literally. Since the cumulative variance only appears to reach about 30% by 156 components, the 50% and 70% outputs are most likely a coding artifact from using argmax() on thresholds that were never actually reached within the first 200 components. So the scree plot still supports the idea of diminishing returns, but it does not support the claim that one component dominates half or more of the variance.

Side-by-Side Recommendations

In [4]:
```
def compare_models(seed_title, seed_artist=None, k=10):
    """Print TF-IDF and LSA recommendations side by side for a seed song."""
    tfidf_recs = tfidf_recommend(
        lyrics_df, tfidf_sim, seed_title=seed_title, seed_artist=seed_artist
    ).rename(columns={"similarity": "tfidf_score"})

    lsa_recs = recommend_songs(
```

```
        lyrics_df, lsa_sim, seed_title=seed_title, seed_artist=seed_artist,
    ).rename(columns={"similarity": "lsa_score"})

    tfidf_recs.index += 1
    lsa_recs.index += 1

    print(f"\nSeed: '{seed_title}'{' by ' + seed_artist if seed_artist else
    print(f"{'Rank':<5} {'— TF-IDF —':<45} {'— LSA —':<45}")
    print('-' * 95)
    for i in range(1, k + 1):
        t = tfidf_recs.loc[i]
        l = lsa_recs.loc[i]
        t_str = f"{t['track_name'][:28]} - {t['artist'][:14]} ({t['tfidf_scc
        l_str = f"{l['track_name'][:28]} - {l['artist'][:14]} ({l['lsa_score
        print(f"{i:<5} {t_str:<45} {l_str:<45}")

    return tfidf_recs, lsa_recs
```

In [5]:
```
# --- Grunge ---
tfidf_grunge, lsa_grunge = compare_models('Iron Clad Lou', seed_artist='Hum'
```

```
Seed: 'Iron Clad Lou' by Hum
Rank  — TF-IDF —                                    — LSA —
-------------------------------------------------------------------------------
-------------------
1     Stand up and Win - Spymob (0.144)             muscles - not dvr (0.51
7)
2     Never Lonely - Space Mountain (0.135)         Never Lonely - Space Mou
ntain (0.504)
3     Downtown - Varsity (0.123)                    Wake up Jacob - Sitcom
(0.504)
4     Rain - World's Experi (0.119)                 Lonely Mess - Laundry
(0.501)
5     Faxing It In - Smushie (0.112)                Hell N Back - Bakar (0.4
88)
6     Hippies - flipturn (0.109)                    Stadium Arcadium - Red H
ot Chili  (0.478)
7     Warmest Part of the Winter - Voxtrot (0.106)  Motorola - Miloe (0.465)
8     2023 - Blue Rain Boot (0.105)                 November - Psychedelic P
o (0.449)
9     Alive - Pearl Jam (0.103)                     Great Dane - Cosmo Pyke
(0.442)
10    Down the Line - Beach Fossils (0.090)         The Green Superintendent
- Mako Road (0.440)
```

The LSA recommender has much higher similarity scores compared to TF-IDF alone. Still, the LSA reommender didn't rank a single grunge song in the top-10, further supporting the argument that grunge music is best represented through its hardcore rock sound.

In [6]:
```
# --- Soul ---
tfidf_soul, lsa_soul = compare_models("Can I Help You?", seed_artist='Amnest
```

```
Seed: 'Can I Help You?' by Amnesty
Rank  ── TF-IDF ──                                  ── LSA ──
--------------------------------------------------------------------------------
-------------------
1     Sweetest Thing on This Side  — Papa Bear & Hi (0.307) Sweetest Thing o
n This Side  — Papa Bear & Hi (0.489)
2     Chicago — Michael Jackso (0.171)              Lean on Me — Bill Wither
s (0.463)
3     Bailee — The Licks (0.161)                    Wizard Of Finance — Parl
iament (0.447)
4     Wizard Of Finance — Parliament (0.149)        Some Are Lakes — Land of
Talk (0.434)
5     Cause = Time — Broken Social  (0.144)         Chicago — Michael Jackso
(0.413)
6     Mystery Girl — Housecall (0.123)              Need — Mustard Servic
(0.405)
7     Ellie — Make Friends (0.118)                  Blue And Sentimental — O
scar Peterson (0.403)
8     Hard To Handle — The Black Crow (0.098)       Simulation — Reptaliens
(0.403)
9     Hey, Who Really Cares — Linda Perhacs (0.095) Chemicals — Rosemary Fai
rw (0.402)
10    Emerald Eyes — Fleetwood Mac (0.091)          Cause = Time — Broken So
cial  (0.395)
```

Again, the LSA recommender his showing significantly higher similarity scores. Additionally, it ranked 3 soul songs in the top-10; one better than the TF-IDF model.

In [7]:
```python
# --- Latin ---
tfidf_latin, lsa_latin = compare_models(
    'Los Libros de la Buena Memoria', seed_artist='Invisible'
)
```

```
Seed: 'Los Libros de la Buena Memoria' by Invisible
Rank  ── TF-IDF ──                                  ── LSA ──
------------------------------------------------------------------------------
-------------------
1     SMUCKERS (feat. Lil Wayne &  — Tyler, The Cre (0.497) La Balsa — Los G
atos (0.973)
2     Las Habladurias del Mundo — Pescado Rabios (0.458) Oh Qué Será? — Will
ie Colón (0.968)
3     After The Storm (feat. Tyler — Kali Uchis (0.445) SMUCKERS (feat. Lil
Wayne &  — Tyler, The Cre (0.966)
4     Como Mata el Viento Norte — La Maquina De  (0.427) Elementales Leches
— Invisible (0.965)
5     Oh Qué Será? — Willie Colón (0.421)           Como Mata el Viento Nort
e — La Maquina De  (0.964)
6     Promesas Sobre El Bidet — Charly García (0.364) Danger (Spider) (Offse
t & JI — Offset (0.959)
7     ·—·· · — ·· —  — —· — — —  — · ·—· ·—  ··· · (0.358) Las Habladurias del
Mundo — Pescado Rabios (0.957)
8     Elementales Leches — Invisible (0.353)        Lo Que Nos Ocupa Es Esa
Abue — Invisible (0.944)
9     La Balsa — Los Gatos (0.316)                  After The Storm (feat. T
yler — Kali Uchis (0.937)
10    G-String — Jv (0.302)                         Muchacha (Ojos de Papel)
— Almendra (0.936)
```

For latin music, the LSA model shows extremely high similarity scores. I expected this because my corpus of songs doesn't have many latin songs compared to other genres and the simple fact of them all being songs in spanish or portugese should make it easy for the LSA to accurate identify that as a topic. Seven out of the top-10 most similar songs were latin in the LSA model compared to only 6 in the TF-IDF model. Not much of a difference on that front, however, the similarity scores are much higher indicating the LSA recommender has a lot more confidence in this genre. It is also important to note that a few hip-hop songs are present in both top-10 rankings. This is most likely due to adlibs being confused for latin words.

In [8]:
```
# --- Hip-hop ---
tfidf_hiphop, lsa_hiphop = compare_models('House Money', seed_artist='Baby K
```

```
Seed: 'House Money' by Baby Keem
Rank  — TF-IDF —                                    — LSA —
--------------------------------------------------------------------------------
-------------------
1     Jesus Is The One (I Got Depr – Zack Fox (0.441) Jesus Is The One (I Go
t Depr – Zack Fox (0.899)
2     Can You Rap Like Me? – Trippie Redd (0.363)   Can You Rap Like Me? – T
rippie Redd (0.834)
3     Hit Bout It – Lil Yachty (0.329)              trademark usa – Baby Kee
m (0.818)
4     Blood On Me – Sampha (0.321)                  REEL IT IN – Aminé (0.81
2)
5     REEL IT IN – Aminé (0.286)                    CORSO – Tyler, The Cre
(0.801)
6     trademark usa – Baby Keem (0.272)             Hit Bout It – Lil Yachty
(0.787)
7     JayJay-Z – Zelooperz (0.266)                  JayJay-Z – Zelooperz (0.
756)
8     Fresh Prince Of Utah – YoungBoy Never (0.262) Fresh Prince Of Utah – Y
oungBoy Never (0.738)
9     CORSO – Tyler, The Cre (0.260)                NDA – Zelooperz (0.721)
10    WILSHIRE – Tyler, The Cre (0.237)             Estaciones [Feat. Hologr
am] – Action Bronson (0.707)
```

Further proving the LSA model's strength, the hip-hop similarity scores are significantly higher than those of TF-IDF. However, both top-10 lists are exclusively hip-hop which has been consistent across all tests. The signficantly higher similarity scores could be due to the dominating adlib topics I discussed previously.

Overlap Analysis

In [9]:
```python
def overlap_stats(tfidf_recs, lsa_recs, label):
    tfidf_set = set(zip(tfidf_recs['track_name'], tfidf_recs['artist']))
    lsa_set   = set(zip(lsa_recs['track_name'],   lsa_recs['artist']))
    shared    = tfidf_set & lsa_set
    jaccard   = len(shared) / len(tfidf_set | lsa_set)
    print(f"{label}")
    print(f"  Shared songs:    {len(shared)}/10")
    print(f"  Jaccard score:   {jaccard:.2f}")
    if shared:
        print(f"  Overlap:         {', '.join([s[0] for s in shared])}")
    print()

overlap_stats(tfidf_grunge,  lsa_grunge,  'Grunge  (Iron Clad Lou)')
overlap_stats(tfidf_soul,    lsa_soul,    'Soul    (Can I Help You?)')
overlap_stats(tfidf_latin,   lsa_latin,   'Latin   (Los Libros...)')
overlap_stats(tfidf_hiphop,  lsa_hiphop,  'Hip-hop (House Money)')
```

```
Grunge  (Iron Clad Lou)
  Shared songs:    1/10
  Jaccard score:   0.05
  Overlap:         Never Lonely

Soul    (Can I Help You?)
  Shared songs:    4/10
  Jaccard score:   0.25
  Overlap:         Sweetest Thing on This Side of Heaven, Chicago, Wizard Of
Finance, Cause = Time

Latin   (Los Libros...)
  Shared songs:    7/10
  Jaccard score:   0.54
  Overlap:         Oh Qué Será?, Como Mata el Viento Norte, La Balsa, SMUCKE
RS (feat. Lil Wayne & Kanye West), Elementales Leches, Las Habladurias del M
undo, After The Storm (feat. Tyler, The Creator & Bootsy Collins)

Hip-hop (House Money)
  Shared songs:    8/10
  Jaccard score:   0.67
  Overlap:         CORSO, Hit Bout It, REEL IT IN, Fresh Prince Of Utah, Can
You Rap Like Me?, JayJay-Z, trademark usa, Jesus Is The One (I Got Depressio
n)
```

Still, no model is doing well with grunge music, however these are by for this highest overlap/similarity scores seen by any model so far. Clearly using LSA to categorize songs by topics imitating lyrical similarity across genres is working quite well.

N-components Test

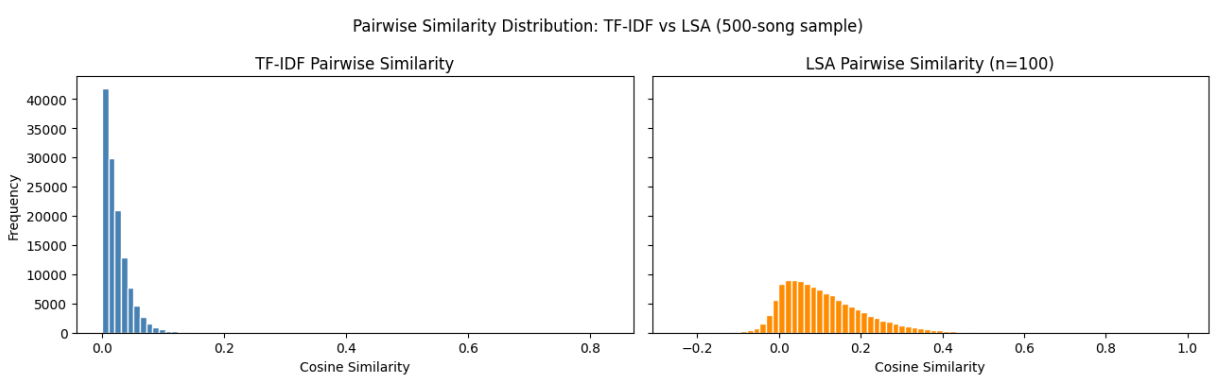
In [10]:
```
results_by_k = {}
for n in [50, 100, 200]:
    _, _, sim_n, _, _ = build_lsa_recommender(LYRICS_PATH, n_components=n)
    recs = recommend_songs(lyrics_df, sim_n, 'House Money', 'Baby Keem', k=1
    results_by_k[n] = set(zip(recs['track_name'], recs['artist']))
    print(f"n={n}: {list(recs['track_name'])[:5]}")

print(f"\nOverlap 50 vs 100:  {len(results_by_k[50]  & results_by_k[100])}/1
print(f"Overlap 100 vs 200: {len(results_by_k[100] & results_by_k[200])}/10"
print(f"Overlap 50 vs 200:  {len(results_by_k[50]  & results_by_k[200])}/10"
```

```
n=50: ['Jesus Is The One (I Got Depression)', 'Hit Bout It', 'CORSO', 'trade
mark usa', 'REEL IT IN']
n=100: ['Jesus Is The One (I Got Depression)', 'Can You Rap Like Me?', 'trad
emark usa', 'REEL IT IN', 'CORSO']
n=200: ['Jesus Is The One (I Got Depression)', 'Can You Rap Like Me?', 'REEL
IT IN', 'trademark usa', 'Hit Bout It']

Overlap 50 vs 100:  8/10
Overlap 100 vs 200: 9/10
Overlap 50 vs 200:  7/10
```

I don't fully get these results.

Similarity Score Distribution: LSA vs TF-IDF

In [11]:
```
sample_size = 500
rng = np.random.default_rng(42)
idx = rng.choice(len(lyrics_df), size=sample_size, replace=False)

lsa_sample   = lsa_sim[np.ix_(idx, idx)]
tfidf_sample = tfidf_sim[np.ix_(idx, idx)]

lsa_upper   = lsa_sample[np.triu_indices(sample_size, k=1)]
tfidf_upper = tfidf_sample[np.triu_indices(sample_size, k=1)]

fig, axes = plt.subplots(1, 2, figsize=(13, 4), sharey=True)

axes[0].hist(tfidf_upper, bins=80, color='steelblue', edgecolor='white')
axes[0].set_title('TF-IDF Pairwise Similarity')
axes[0].set_xlabel('Cosine Similarity')
axes[0].set_ylabel('Frequency')

axes[1].hist(lsa_upper, bins=80, color='darkorange', edgecolor='white')
axes[1].set_title(f'LSA Pairwise Similarity (n={N_COMPONENTS})')
axes[1].set_xlabel('Cosine Similarity')

plt.suptitle('Pairwise Similarity Distribution: TF-IDF vs LSA (500-song samp
plt.tight_layout()
plt.show()

for label, arr in [('TF-IDF', tfidf_upper), ('LSA', lsa_upper)]:
    print(f"{label}:  mean={arr.mean():.4f}  median={np.median(arr):.4f}  "
          f">0.1: {(arr > 0.1).mean()*100:.1f}%  >0.3: {(arr > 0.3).mean()*1
```




```
TF-IDF:  mean=0.0233  median=0.0172  >0.1: 1.2%  >0.3: 0.1%
LSA:  mean=0.1099  median=0.0896  >0.1: 45.9%  >0.3: 5.4%
```

LSA scores are much higher on average and more normally distributed. TF-IDF scores cluster near zero because exact vocabulary overlap is rare. This distribution shift is one reason LSA sometimes produces better recommendatons - it can find connections TF-IDF misses - but also why it can over connected unrelated songs.

Dominant Topics Per Seed Song

In [12]:
```
seeds = [
    ('Iron Clad Lou',                 'Hum'),
    ("Can I Help You?",               'Amnesty'),
    ('Los Libros de la Buena Memoria','Invisible'),
    ('House Money',                   'Baby Keem'),
]

for title, artist in seeds:
    try:
        topic_scores = get_song_top_topics(
            lyrics_df, lsa_matrix, title, seed_artist=artist, n_topics=5
        )
        print(f"'{title}' — dominant topics:")
        print(topic_scores.to_string())
        print()
    except ValueError as e:
        print(f"[NOT FOUND] {e}\n")
```

```
'Iron Clad Lou' — dominant topics:
Topic 1     0.4066
Topic 32    0.2457
Topic 79    0.2036
Topic 88    0.1914
Topic 80   -0.1797

'Can I Help You?' — dominant topics:
Topic 44   -0.2933
Topic 85   -0.2521
Topic 1     0.2402
Topic 55   -0.2233
Topic 26    0.2122

'Los Libros de la Buena Memoria' — dominant topics:
Topic 11    0.6511
Topic 2     0.5081
Topic 12   -0.3827
Topic 13   -0.1577
Topic 14   -0.1384

'House Money' — dominant topics:
Topic 4     0.4807
Topic 1     0.3820
Topic 8     0.2675
Topic 5     0.2513
Topic 31    0.1842
```

Topic one was the strongest in grunge by far. It includes tokens like: im, dont, know, love, oh, yeah. It has a bit of a emo feeling to it, but still not very descriptive. Soul music struggled with most topics it seems, which is odd to me becasue I associate soul music with some clear cultural messaging. Latin music is heavily dominated by topics 11 and 2

which isn't surprising because they are comprised of mainly spanish words not frequent in the rest of the corpus outside of other latin songs. Hip-hop had the most even distribution across topics which is further validating my hypothesis that adlib dominante songs are heavily influencing the topics creating by LSA. Hip-hop songs in general follow very similar pattern in both sound and lyrical messaging. They also include the most explicatives by far.

10. Observations & Comparison Notes

LSA improved on TF-IDF by making the similarity space much denser and less dependent on exact word matching. Compared with TF-IDF, its pairwise similarities were much higher on average, which suggests the model was able to connect songs through broader word co-occurrence patterns rather than only direct overlap. That is the clearest strength of the LSA approach. At the same time, the latent topics still showed major overlap in adlibs, repeated emotional language, and filler phrases, so the model did not fully escape the structure of the original lyric vocabulary.

LSA often feels more flexible than TF-IDF, but that flexibility can come at the cost of precision. Since songs are projected into a compressed latent space, some recommendations become more semantically broad but also more weakly grounded in clearly interpretable lyric evidence. In other words, LSA helps solve the sparsity problem of TF-IDF, but it can also connect songs that are not truly close matches. For this project, that makes LSA a meaningful improvement over the baseline, but not yet a fully satisfying solution.